1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlighted that non-invasive urine methylation, tumor mutational/neoantigen load (bTMB) and immune-contexture (Immunoscore) converge but do not fully explain why some MIBC tumors with apparently favorable immune contexture fail to respond to immunotherapy. The core question is: under what tumor-intrinsic or microenvironmental conditions do organized lymphoid structures (or high local T-cell content) fail to translate into checkpoint-blockade sensitivity?

Working Hypothesis

Primary hypothesis: In muscle-invasive bladder cancer (MIBC), TLS-like lymphoid aggregates reflect organized intratumoral immunity, but their anti-tumor potential is impeded when tumor-derived S100A9 skews myeloid cells toward a PD-L1–high, T cell–suppressive phenotype; accordingly, TLS-high tumors with high S100A9 and PD-L1 will show inferior response to PD-(L)1 blockade compared with TLS-high tumors with low S100A9. The S100A9–PD-L1 axis represents a targetable resistance mechanism, and modulating S100A9 signaling or PD-L1 within TLS-rich microenvironments should restore TLS-driven T cell activity and improve immunotherapy response. Secondary modifiers such as TP53 status or metabolic cues will tune the magnitude of this effect.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	S100A9, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

S100A9: S100A9 and PD-L1 as anchors because Tier 1 paths specifically implicated S100A9 in modulating PD-L1 and immunosuppression.

PD-L1: PD-L1 and Immunoscore are used clinically to triage immunotherapy, making response the natural primary endpoint.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	bTMB	Supporting modifier from knowledge graph
Secondary	urine-DNA methylation biomarkers	Supporting modifier from knowledge graph

Secondary	TP53	Supporting modifier from knowledge graph
Secondary	Immunoscore	Supporting modifier from knowledge graph
Secondary	ALP	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	TG	Supporting modifier from knowledge graph
Secondary	SHBG	Supporting modifier from knowledge graph
Secondary	IGF-1	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective and prospective cohorts to correlate TLS density/maturation and S100A9/PD-L1 expression within TLS regions with objective response to PD-(L)1 therapy;

Prediction 2: multiplex imaging and spatial transcriptomics to map S100A9 and PD-L1 within TLS microdomains;

Prediction 3: ex vivo TLS organoid/co-culture systems or humanized mouse models to test effects of S100A9 inhibition or PD-L1 blockade on T cell activation in TLS-rich contexts;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- urine-DNA methylation biomarkers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.